

SHINYMR: A SHINY APP FOR RIGOROUS AND REPRODUCIBLE MENDELIAN RANDOMIZATION

Joint Statistical Meetings

Fred Boehm

Monday, August 4, 2025

COLLABORATIVE RESEARCH WITH JI HOON PARK



WHAT IS A SNP?

Individual 1

Maternal ...CGATATTCC**T**ATCGAATGTC...

Paternal ...CGATATTCC**C**ATCGAATGTC...

Individual 2

Maternal ...CGATATTCC**C**ATCGAATGTC...

Paternal ...CGATATTCC**C**ATCGAATGTC...

Individual 3

Maternal ...CGATATTCC**T**ATCGAATGTC...

Paternal ...CGATATTCC**T**ATCGAATGTC...

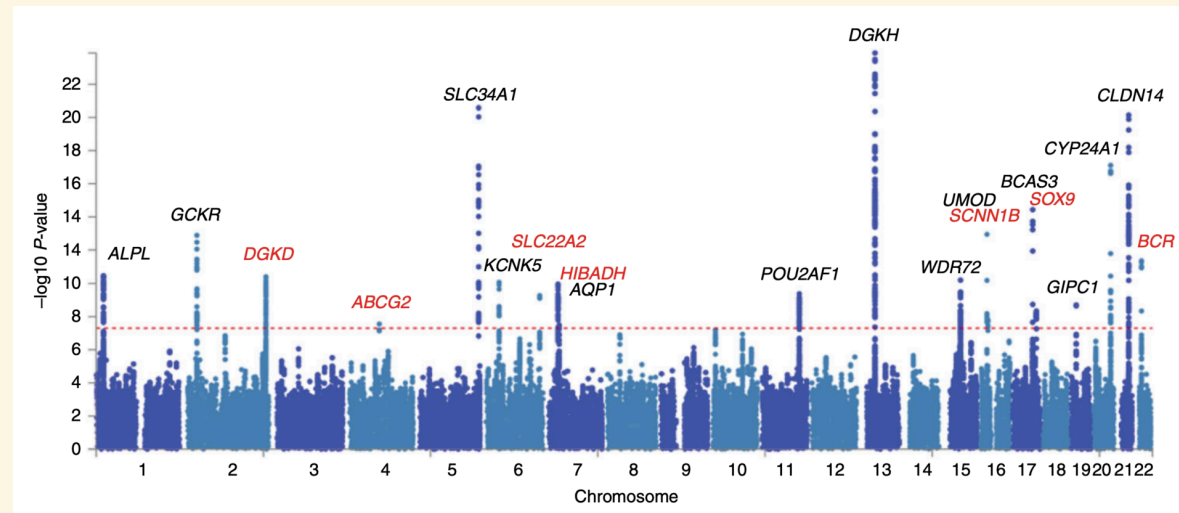
Individual 4

Maternal ...CGATATTCC**C**ATCGAATGTC...

Paternal ...CGATATTCC**T**ATCGAATGTC...

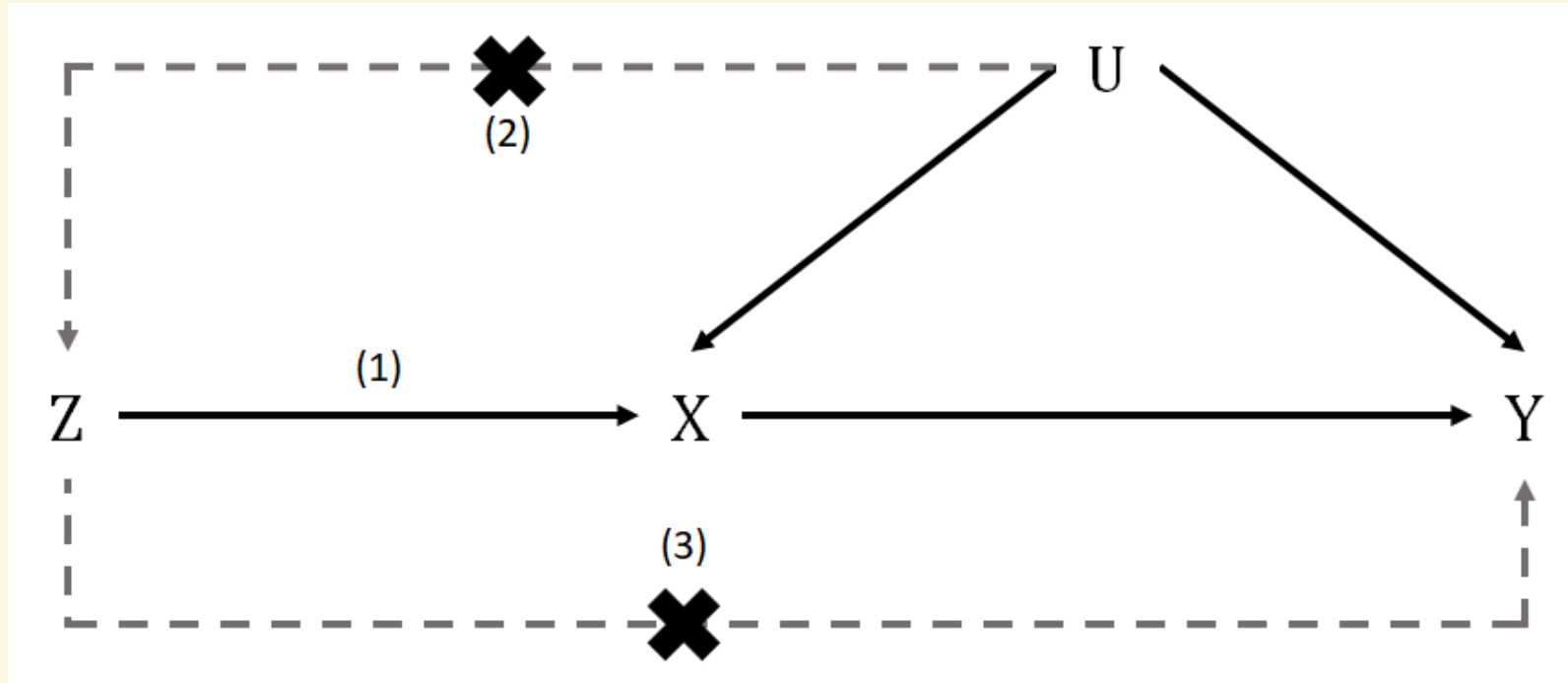
GENOME-WIDE ASSOCIATION STUDIES

- A study to detect genotype-phenotype associations
- Involves thousands of subjects with SNP genotypes and phenotypes
- GWAS results per SNP are publicly shared



GWAS for kidney stone disease

WHAT IS MENDELIAN RANDOMIZATION?



- Instrumental variable analysis to characterize evidence for a causal effect of the “exposure” (X) on the “outcome” (Y)

MOTIVATION FOR A SHINY APP

- Many recent reports have questioned the scientific rigor of published MR studies
- To promote rigor and reproducibility, shinyMR users automatically perform MR analysis and sensitivity analyses to evaluate the robustness of MR assumptions

WHAT IS SHINY?



- An R package that enables users to build an interactive webpage “App”
- App can then be deployed on a website for other scientists to use

SHINY APP DESIGN CONSIDERATIONS

- Interface with openGWAS database
- Require sensitivity analyses to assess MR assumptions
- Users must manually ensure that there is no subject overlap in the two GWAS

DEVELOPMENT VERSION OF THE APP

CHOOSING GWASS FROM OPENGWAS DATABASE

Two-sample Mendelian Randomization (MR)

Exposure Trait Keyword:
total triglycerides

Outcome Trait Keyword:
myocardial infarction

Download Analysis Code

TraitsExposure DataOutcome DataMR ResultsPlotSensitivity Analyses

Showing 1 to 3 of 3 entries

Search:

Showing 1 to 3 of 3 entries

Showing 1 to 3 of 3 entries

	id	trait	coverage	ncase	group_name	year	mr	author	sex	qc_prior_to_upload	pmid	priority	population	unit	nsnp	sample_size	build	ncontrol	covariat
1	ebi-a-GCST90092992	Total triglycerides levels			public	2022	1	Richardson TG	NA		35213538		European		11590399	115082	HG19/GRCh37		
2	met-c-934	Serum total triglycerides			public	2016	1	Kettunen	Males and Females		27005778	1	European	SD	11871391	21545	HG19/GRCh37		
3	met-d-Total_TG	Total triglycerides			public	2020	1	Borges CM	Males and Females			0	European	NA	12321875	115078	HG19/GRCh37		
Showing 1 to 3 of 3 entries																			
	id	trait	coverage	ncase	group_name	year	mr	author	sex	qc_prior_to_upload	pmid	priority	population	unit	nsnp	sample_size	build	ncontrol	covariat
1	ebi-a-GCST90038610	Myocardial infarction		11081	public	2021	1	D<U+00F6>nerta<U+015F>HM	NA		33959723		NA		9587836	484598	HG19/GRCh37	4	
2	ebi-a-GCST90018877	Myocardial infarction		20917	public	2021	1	Sakaue S	NA		34594039		European		24172914	461823	HG19/GRCh37	4	
3	ebi-a-GCST90018657	Myocardial infarction		14992	public	2021	1	Sakaue S	NA		34594039		East Asian		12452043	161206	HG19/GRCh37	1	
4	ukb-e-I21_CSA	I21 Acute myocardial infarction		374	public	2020	1	Pan-UKB team	Males and Females				0	South Asian	NA	9805094	8876	HG19/GRCh37	
5	ukb-d-I9_MI	Myocardial infarction		7018	public	2018	1	Neale lab	Males and Females				0	European	NA	12640541	361194	HG19/GRCh37	3
6	ukb-d-I9_MI_STRICT	Myocardial infarction, strict		7018	public	2018	1	Neale lab	Males and Females				0	European	NA	12640541	361194	HG19/GRCh37	3
7	ukb-b-6295	Diagnoses - secondary ICD10: Z03.4 Observation for suspected myocardial infarction		2559	public	2018	1	Ben Elsworth	Males and Females				1	European	SD	9851867	463010	HG19/GRCh37	4
8	ukb-b-5126	Diagnoses - main ICD10: I21.1 Acute transmural myocardial		1673	public	2018	1	Ben Elsworth	Males and Females				1	European	SD	9851867	463010	HG19/GRCh37	4

127.0.0.1:4492/#tab-5311-1

EXTRACT SNP INSTRUMENTS

TraitsExposure DataOutcome DataMR ResultsPlotSensitivity Analyses

Show10▼entries

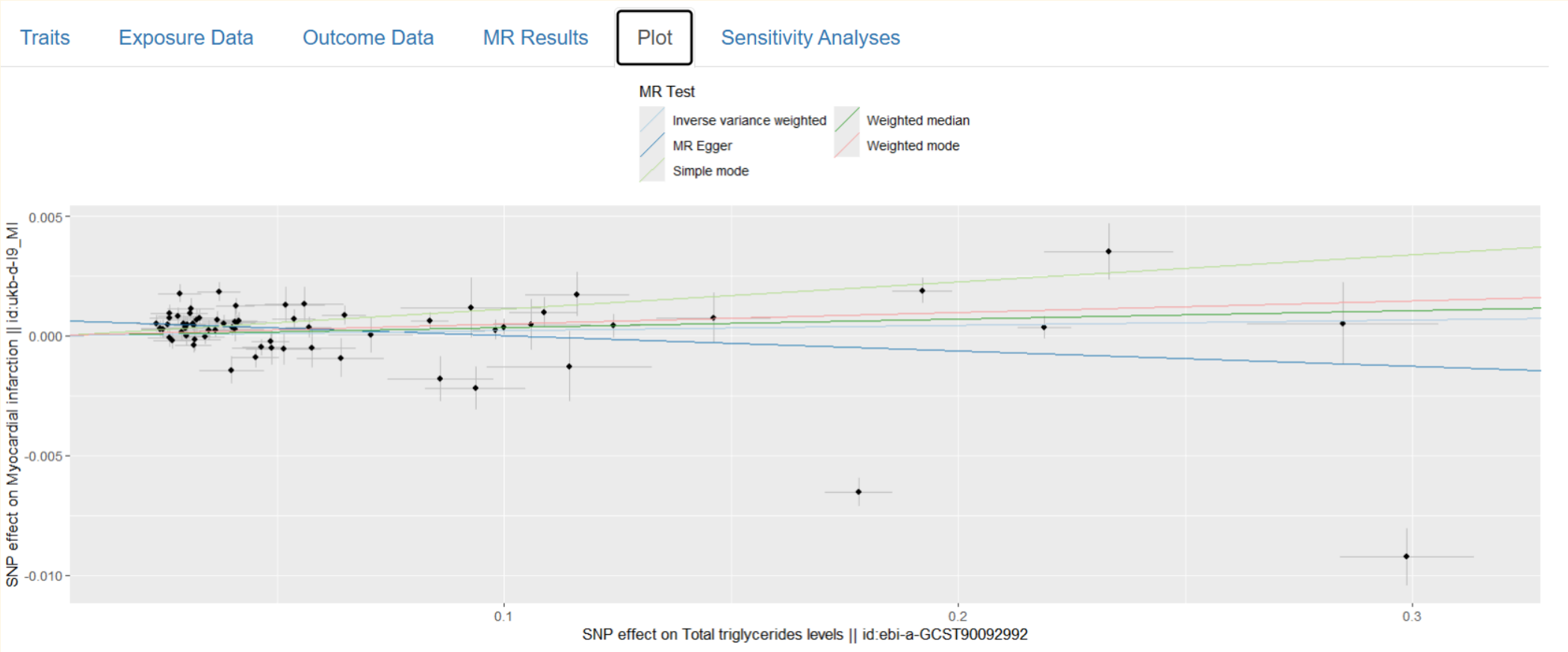
Search:

	id.exposure	chr.exposure	pos.exposure	SNP	effect_allele.exposure	other_allele.exposure	eaf.exposure	beta.exposure	se.exposure	pval.exposure	samplesize.exposure	exposure	mr_keep.exposure	pval_origin.exposure	data_source.exposure
1	ebi-a-GCST90092992	11	118648917	rs984184	C	G	0.86728	-0.218989	0.00588514	1e-200	115082	Total triglycerides levels id:ebi-a-GCST90092992	true	reported	igd
2	ebi-a-GCST90092992	8	19819724	rs328	G	C	0.100307	-0.192009	0.0095995	4.19759e-188	115082	Total triglycerides levels id:ebi-a-GCST90092992	true	reported	igd
3	ebi-a-GCST90092992	2	27730940	rs1280326	C	T	0.803982	-0.100041	0.00406473	9.39723e-134	115082	Total triglycerides levels id:ebi-a-GCST90092992	true	reported	igd
4	ebi-a-GCST90092992	6	161010118	rs10455872	G	A	0.078987	-0.178066	0.00736873	5.19996e-129	115082	Total triglycerides levels id:ebi-a-GCST90092992	true	reported	igd
5	ebi-a-GCST90092992	8	126500031	rs28601781	G	C	0.420159	-0.0863904	0.00407457	9.09913e-100	115082	Total triglycerides levels id:ebi-a-GCST90092992	true	reported	igd
6	ebi-a-GCST90092992	7	73020337	rs3812316	G	C	0.129282	-0.124142	0.00592726	2.09991e-97	115082	Total triglycerides levels id:ebi-a-GCST90092992	true	reported	igd
7	ebi-a-GCST90092992	6	180622870	rs117733303	G	A	0.0185	-0.298893	0.0147465	3.19963e-91	115082	Total triglycerides levels id:ebi-a-GCST90092992	true	reported	igd
8	ebi-a-GCST90092992	1	63060037	rs70439217	TAAGAA	T	0.352737	-0.08371	0.00416345	6.4998e-90	115082	Total triglycerides levels id:ebi-a-GCST90092992	true	reported	igd
9	ebi-a-GCST90092992	2	21226560	rs2678379	G	A	0.793822	0.0980421	0.00490244	5.70033e-89	115082	Total triglycerides levels id:ebi-a-GCST90092992	true	reported	igd
10	ebi-a-GCST90092992	19	45430280	rs5112	G	C	0.533735	0.0801811	0.00428077	2.80027e-78	115082	Total triglycerides levels id:ebi-a-GCST90092992	true	reported	igd

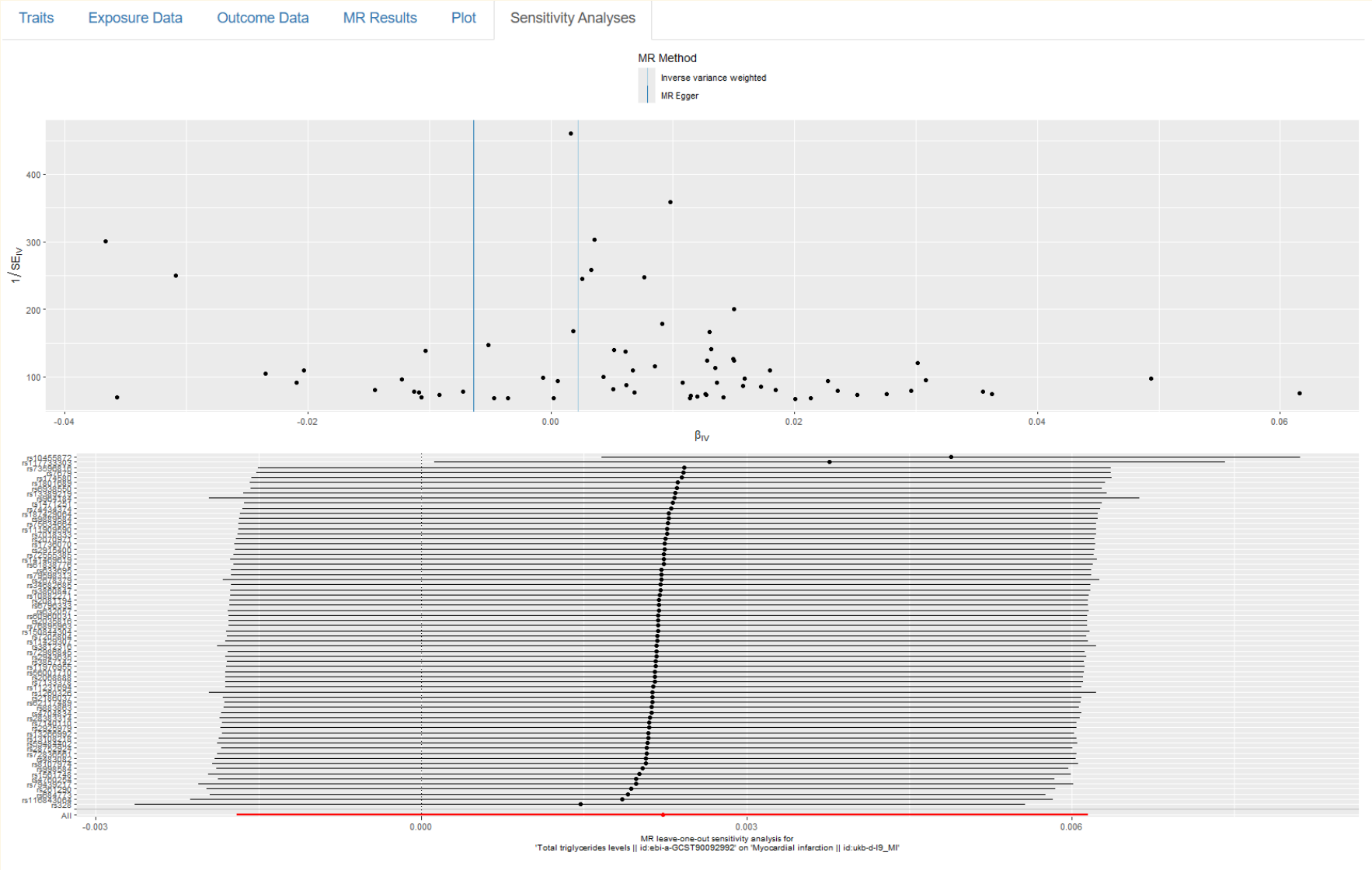
Showing 1 to 10 of 72 entries

Previous12345...8Next

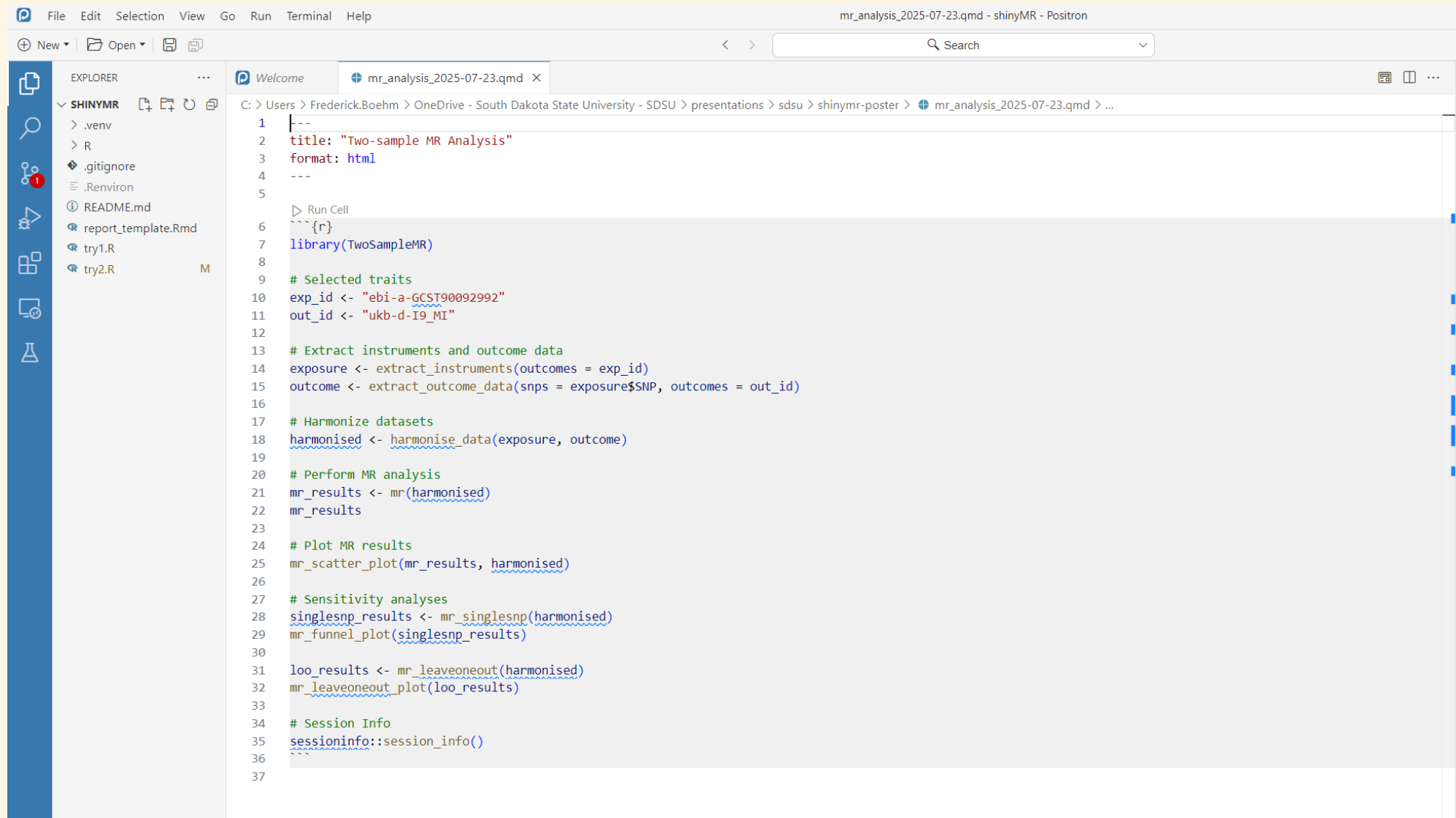
PERFORM MR ANALYSIS



PERFORM SENSITIVITY ANALYSES



DOWNLOAD A REPRODUCIBLE QUARTO DOCUMENT



The screenshot displays the Positron IDE interface. The top menu bar includes File, Edit, Selection, View, Go, Run, Terminal, and Help. The title bar shows the file name: mr_analysis_2025-07-23.qmd - shinyMR - Positron. The Explorer panel on the left shows the project structure for SHINYMR, including files like .venv, R, .gitignore, .Renvirom, README.md, report_template.Rmd, try1.R, and try2.R. The main editor area shows the content of mr_analysis_2025-07-23.qmd, which is a Quarto document. The document includes a title, format, and a series of R code blocks for MR analysis.

```
1  |---
2  |title: "Two-sample MR Analysis"
3  |format: html
4  |---
5
6  | Run Cell
7  |{r}
8  |library(TwoSampleMR)
9
10 | # Selected traits
11 | exp_id <- "ebi-a-GCST90092992"
12 | out_id <- "ukb-d-I9_MI"
13
14 | # Extract instruments and outcome data
15 | exposure <- extract_instruments(outcomes = exp_id)
16 | outcome <- extract_outcome_data(snps = exposure$SNP, outcomes = out_id)
17
18 | # Harmonize datasets
19 | harmonised <- harmonise_data(exposure, outcome)
20
21 | # Perform MR analysis
22 | mr_results <- mr(harmonised)
23
24 | # Plot MR results
25 | mr_scatter_plot(mr_results, harmonised)
26
27 | # Sensitivity analyses
28 | singlesnp_results <- mr_singlesnp(harmonised)
29 | mr_funnel_plot(singlesnp_results)
30
31 | loo_results <- mr_leaveoneout(harmonised)
32 | mr_leaveoneout_plot(loo_results)
33
34 | # Session Info
35 | sessioninfo::session_info()
36
37
```

THANK YOU!